

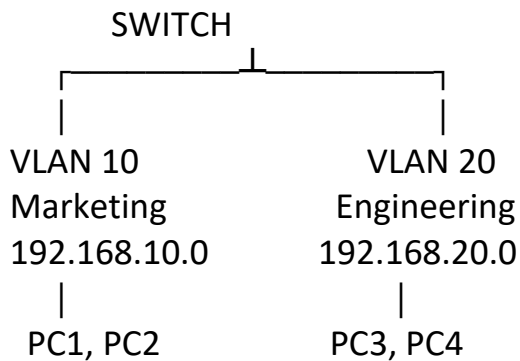
Question: Explain the purpose of VLAN segmentation, and describe how an 802.1Q trunk link allows traffic for multiple VLANs to travel between the switch and the router over a single physical connection.

In this startup, **Marketing** and **Engineering** use the same physical switch, but we want their traffic to be **logically separated**.

1. What is VLAN segmentation?

A **VLAN (Virtual LAN)** divides one physical switch into multiple logical networks.

Example:



Even though all PCs are connected to the **same physical switch**, they belong to different logical networks.

Why use VLANs?

1. Security / Isolation

Marketing devices don't automatically belong to the same Layer-2 network as Engineering devices.

2. Broadcast control

A broadcast from Marketing stays within **VLAN 10** instead of reaching Engineering's VLAN 20.

3. Organization

You can easily assign:

- VLAN 10 → Marketing
- VLAN 20 → Engineering

4. Better network management

Each department can have its own IP subnet and security policies.

2. What is an 802.1Q trunk?

Now suppose Marketing and Engineering need to communicate with other networks through **one router**.

Instead of using two separate physical cables:

Switch — Router ← Marketing
Switch — Router ← Engineering

we can use **one physical connection**:

802.1Q TRUNK
Switch ===== Router
VLAN 10 + VLAN 20

The trunk carries traffic belonging to **multiple VLANs over the same physical link**.

3. How does the router know which VLAN traffic belongs to?

This is where **802.1Q tagging** is used.

When a frame travels across the trunk, an **802.1Q VLAN tag** identifies its VLAN.

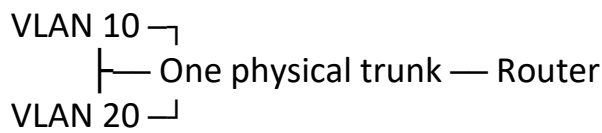
For example:

Marketing frame
↓
[Ethernet Frame | VLAN 10 Tag]
↓
TRUNK
↓
Router

And:

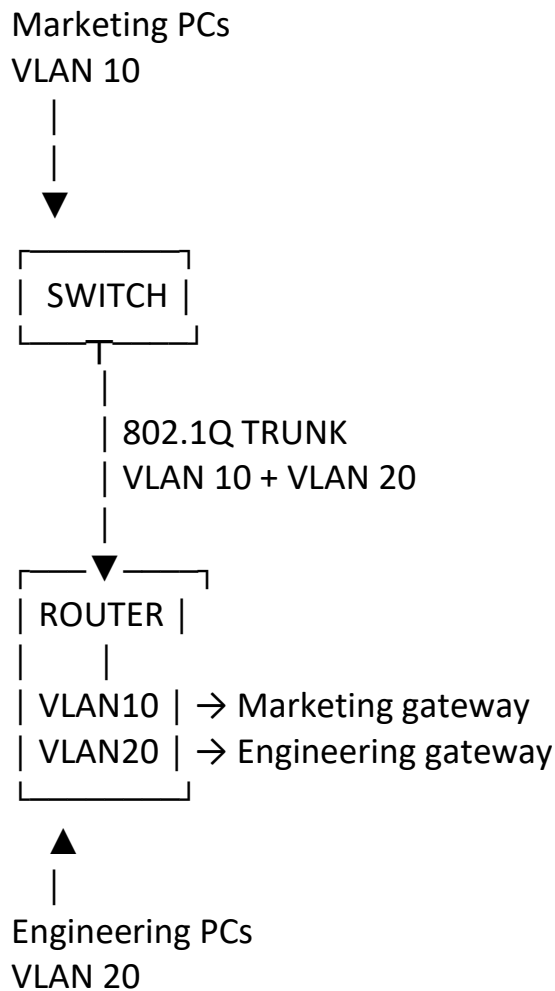
Engineering frame
↓
[Ethernet Frame | VLAN 20 Tag]
↓
TRUNK
↓
Router

So the same cable can carry:



4. Router-on-a-Stick

A common setup for this scenario is called **Router-on-a-Stick**.



The router uses **subinterfaces** such as:

G0/0.10 → VLAN 10
G0/0.20 → VLAN 20

This allows the router to perform **inter-VLAN routing**.

VLAN segmentation logically separates departments on the same physical switch. For example, Marketing can use VLAN 10 and Engineering can use VLAN 20. This provides logical isolation and reduces unnecessary broadcast traffic. An 802.1Q trunk allows multiple VLANs to travel over one physical link between the switch and router by adding VLAN tags to Ethernet frames. The router can use separate subinterfaces for each VLAN and perform inter-VLAN routing.